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NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Mesothelioma: Pleural

Version 3.2026 — July 8, 2026

NCCN Guidelines for Patients®

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National Comprehensive Cancer Network®

NCCN Guidelines Version 3.2026
Mesothelioma: Pleural

NCCN Guidelines Index
Table of Contents
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National Comprehensive Cancer Network®

NCCN Guidelines Version 3.2026
Mesothelioma: Pleural

NCCN Guidelines Index
Table of Contents
Discussion

NCCN Mesothelioma: Pleural Panel Members
Summary of the Guidelines Updates

Initial Evaluation (PM-1)
Pretreatment Evaluation (PM-2)
Clinical Stage I and Epithelioid Histology; Non-Surgical Treatment (PM-2)
Clinical Stage II–IV and Epithelioid Histology; Sarcomatoid or Biphasic Histology (any stage), or Medically Inoperable (any stage); Treatment (PM-2)
Clinical Stage I and Epithelioid Histology; Surgical Evaluation and Treatment (PM-3)
Principles of Pathologic Review (PM-A)
Principles of Supportive Care and Survivorship (PM-B)
Principles of Systemic Therapy (PM-C)
Principles of Radiation Therapy (PM-D)
Principles of Surgery (PM-E)

Staging (ST-1)

Abbreviations (ABBR-1)

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See NCCN Categories of Evidence and Consensus.

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See NCCN Categories of Preference.

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National Comprehensive Cancer Network®

NCCN Guidelines Version 3.2026
Mesothelioma: Pleural

NCCN Guidelines Index
Table of Contents
Discussion

Updates in Version 3.2026 of the NCCN Guidelines for Mesothelioma: Pleural include:

MS-1
- The discussion section has been updated to reflect the changes in the algorithm.

Updates in Version 2.2026 of the NCCN Guidelines for Mesothelioma: Pleural include:

PM-C 1 of 3
- Footnote d added: Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.

Updates in Version 1.2026 of the NCCN Guidelines for Mesothelioma: Pleural include:

PM-1
- Initial Evaluation
  - Bullet added: Assess for distress
- Footnote b added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).

PM-B
- Title change: PRINCIPLES OF SUPPORTIVE CARE *AND SURVIVORSHIP*
- Bullets added for the following NCCN Guidelines
  - Cancer-Related Fatigue
  - Management of Immunotherapy-Related Toxicities
  - Cancer-Associated Venous Thromboembolic Disease
  - Survivorship

PM-C 1 of 3
- Epithelioid; Biphasic or Sarcomatoid
  - Subsequent Therapy
    - ◊ Pemetrexed moved from Preferred to Other Recommended

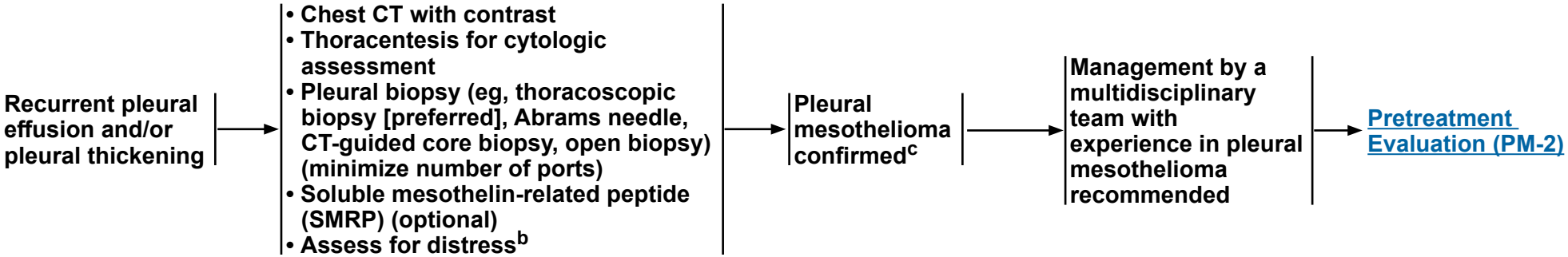
PM-C 2 of 3
- Reference 9 updated.

ST-1 and ST-2
- Table 1: TNM definitions updated as per the 9th edition of the AJCC Cancer Staging System
- Table 2: AJCC Prognostic Groups updated as per the 9th edition of the AJCC Cancer Staging System

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INITIAL EVALUATION[a]



[a] There are no data to suggest that screening improves survival.
[b] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
[c] Principles of Pathologic Review (PM-A).

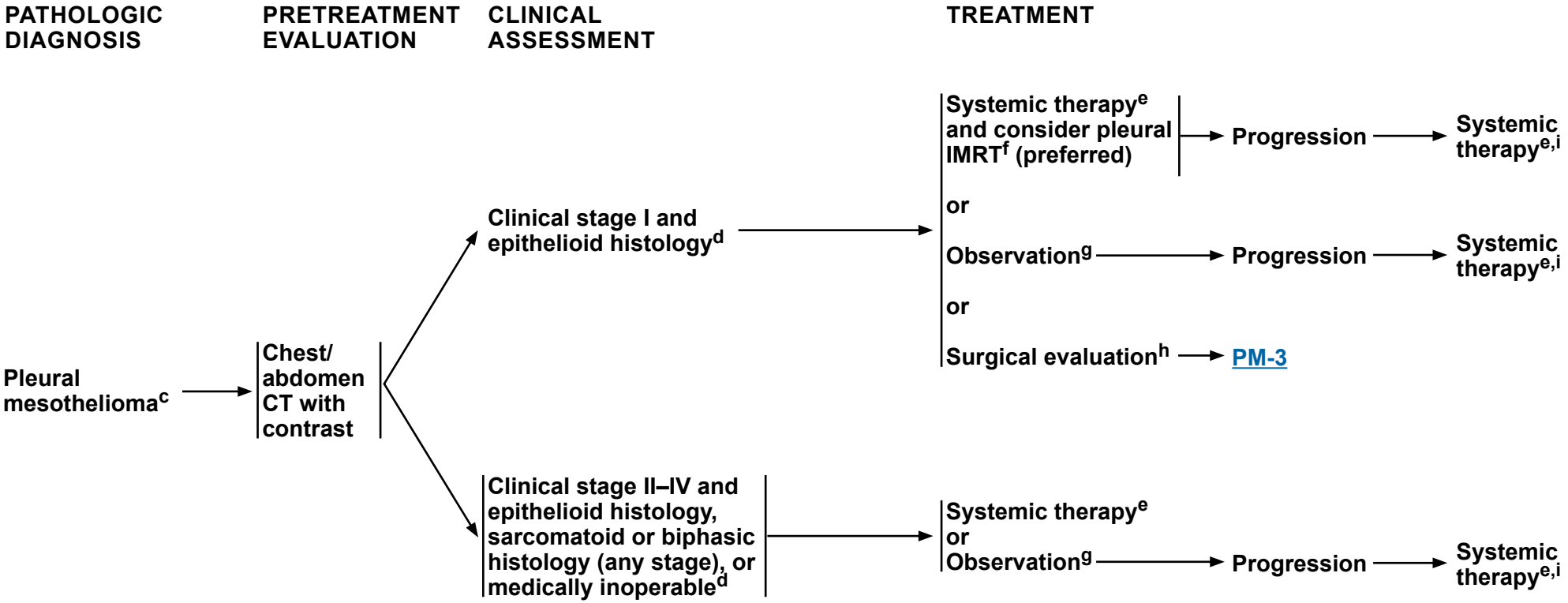
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NCCN Guidelines Version 3.2026
Mesothelioma: Pleural

NCCN Guidelines Index
Table of Contents
Discussion



[c] Principles of Pathologic Review (PM-A).
[d] Best supportive care is recommended for patients presenting with performance status (PS) 3–4. See Principles of Supportive Care and Survivorship (PM-B).
[e] Principles of Systemic Therapy (PM-C).
[f] Pleural intensity-modulated radiation therapy (IMRT) should only be performed in carefully selected patients at centers with experience in this technique. See Principles of Radiation Therapy (PM-D).
[g] Observation may be considered for patients who are asymptomatic with minimal burden of disease if systemic therapy is planned at the time of symptomatic or radiographic progression.
[h] The benefit of surgical resection in pleural mesothelioma is unclear and there is no evidence that patient survival is improved with surgery when combined with systemic therapy versus systemic therapy alone.
[i] Principles of Supportive Care and Survivorship (PM-B).

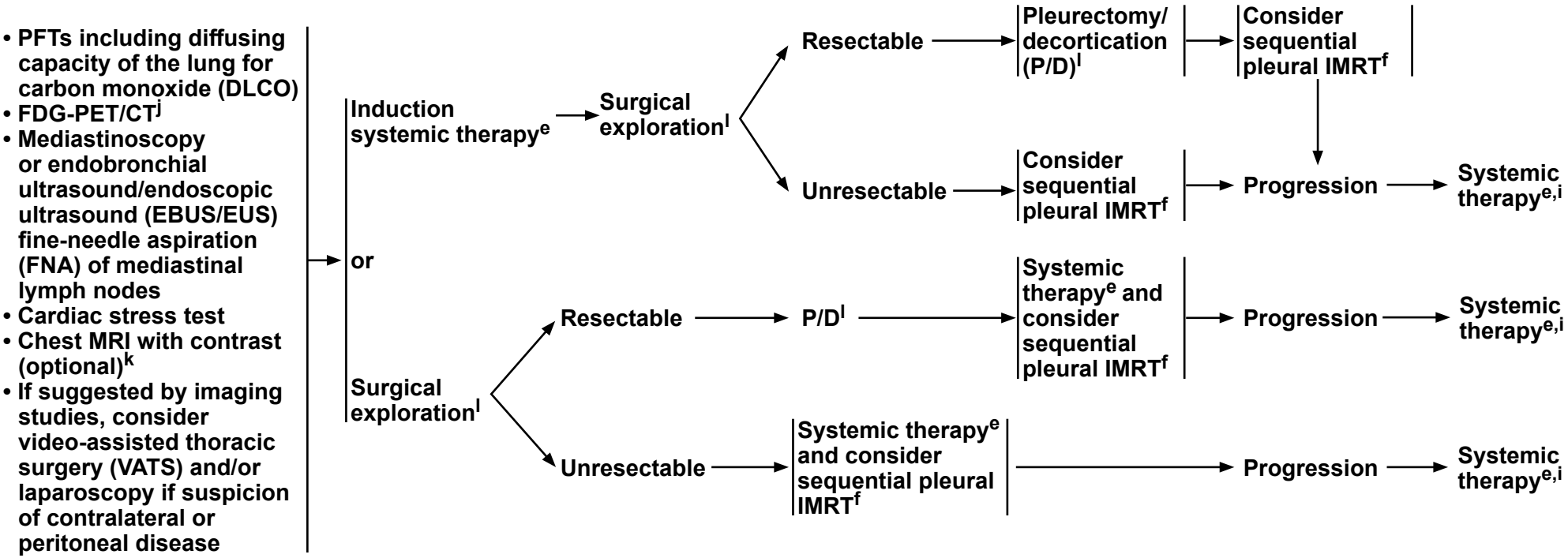
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CLINICAL STAGE I AND EPITHELIOID HISTOLOGY SURGICAL EVALUATION[h]

TREATMENT[i]



[e] Principles of Systemic Therapy (PM-C).
[f] Pleural IMRT should only be performed in carefully selected patients at centers with experience in this technique. See Principles of Radiation Therapy (PM-D).
[h] The benefit of surgical resection in pleural mesothelioma is unclear and there is no evidence that patient survival is improved with surgery when combined with systemic therapy versus systemic therapy alone.
[i] Principles of Supportive Care and Survivorship (PM-B).
[j] If FDG-PET/CT is to be done, recommend obtaining FDG-PET/CT before pleurodesis. Confirm diagnosis of pleural mesothelioma prior to pleurodesis. If pleural mesothelioma is suspected, consider evaluation by a multidisciplinary team with expertise in pleural mesothelioma.
[k] For further evaluation of possible chest, spinal, diaphragmatic, or vascular involvement based on CT imaging.
[l] Principles of Surgery (PM-E).

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PRINCIPLES OF PATHOLOGIC REVIEW

Pathologic Evaluation

- Mesothelioma originates from the cells in the serosal lining that surrounds the body cavities. Of all mesotheliomas, ~85% arise from the pleura, ~15% arise from the peritoneum, and the remainder (<1%) originate from the pericardium or the tunica vaginalis.[1]
- In the United States, diffuse pleural mesothelioma affects ~3,000 patients each year, with an annual incidence of ~1 in 100,000.[2,3]
- The purpose of the pathologic evaluation of mesothelioma is based on the pathologic assessment of tumor tissue, which can be obtained from core biopsy sampling, pleurectomy, or other more extensive resections such as extrapleural pneumonectomy (EPP). Given its rarity and overlapping microscopic features with other conditions, the histologic diagnosis of diffuse mesothelioma can be challenging.
- To establish a pathologic diagnosis of mesothelioma, diagnostic tools that are used clinically include histologic assessment, immunohistochemistry (IHC), cytogenetics, and molecular techniques (eg, targeted next-generation sequencing [NGS], fluorescence in situ hybridization [FISH], single-nucleotide polymorphism arrays). Despite the multiple diagnostic toolkits, the diagnosis relies primarily on proper histologic assessment and IHC.
- The new edition of the World Health Organization (WHO) Classification of Thoracic Tumors by the International Agency for Research on Cancer (IARC) introduced the following changes for 2021 from the previous 2015 edition[1,4]:
  - New entity: mesothelioma in situ
  - New terminology: diffuse pleural mesothelioma (instead of diffuse *malignant* pleural mesothelioma)
  - New terminology: localized pleural mesothelioma (instead of localized *malignant* pleural mesothelioma)
  - New terminology: well-differentiated papillary mesothelial tumor (WDPMT, instead of well-differentiated papillary mesothelioma)
  - Genetic tumor syndromes involving the thorax: *BAP1* tumor predisposition syndrome is a hereditary cancer syndrome caused by heterozygous germline pathogenic variants in the *BAP1* gene.
- The descriptions below refer to diffuse mesothelioma, which will be named *mesothelioma* for the purpose of simplicity.

Mesothelioma Classification

- Mesothelioma is classified into three histologic types: epithelioid, biphasic (mixed), and sarcomatoid, which have significant prognostic value.[1]
- The determination of histologic types is based on the cytologic features of the tumor:
  - *Epithelioid mesothelioma* is characterized by epithelioid-to-round cells.
  - *Sarcomatoid mesothelioma* is characterized by spindled cells with tapered nuclei.
  - *Biphasic mesothelioma* contains both epithelioid and sarcomatoid components in various proportions, with each comprising at least 10% of the tumor.

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PRINCIPLES OF PATHOLOGIC REVIEW

Mesothelioma Classification (continued)

- Within each histologic type, mesothelioma can be divided into several subtypes and patterns based on its cytologic, architectural, and background stromal features.[5]
  - Other rare variants of epithelioid mesothelioma include clear cell, signet ring cell, rhabdoid, deciduoid, and small cell.[6-8] Tumor cells are arranged in diverse architectural patterns that include tubulopapillary, trabecular, solid, acinar, micropapillary, or adenomatoid.
  - In sarcomatoid mesothelioma, subtypes described include conventional/spindle cell, desmoplastic,[9,10] and lymphohistiocytoid.[11-13] A subset of sarcomatoid mesothelioma exhibits heterologous differentiation with osteosarcomatous, chondrosarcomatous, and/or rhabdomyosarcomatous elements.[10]
  - The assignment of histologic type can be challenging, given the intertumoral and intratumoral morphologic heterogeneity. Appropriate type classification of mesothelioma is nonetheless important, given the prognostic significance of different histologic types.
  - Studies comparing the concordance between histologic type in initial biopsies with subsequent resections have shown that the accuracy of typing increases with a higher number of biopsies.[14] While sarcomatoid histology in biopsies is highly predictive of sarcomatoid histology in resections, epithelioid histology in biopsies is not entirely specific and is changed to biphasic or sarcomatoid types in resections in up to 20% of patients.[14]

Histologic Criteria for Mesothelioma

- In mesothelioma, the goals of histologic assessment are to confirm the pathologic diagnosis and to determine the histologic type, which allows for prognostication and treatment planning. For the diagnosis of mesothelioma, one needs to establish each of the three conditions below:
  - ***The lesion is diffuse and not solitary***. Correlation with clinical and radiologic findings is needed to confirm that the distribution of the tumor is diffuse rather than solitary. While almost all (>99%) mesotheliomas are diffuse, rare cases of *localized pleural mesothelioma* have been described, which are solitary, have a different pathogenesis, and harbor a relatively less aggressive clinical course.[15-18]
  - ***The lesional cells are mesothelial***. Given the morphologic overlap between mesothelioma and diverse mimics such as carcinomas, IHC can be used to confirm the presence of mesothelial differentiation in the tumor cells. Other tools such as cytogenetics and molecular analysis may also be helpful in some instances (see next page).
  - ***The lesional cells are malignant***. Histologic assessment is integral to establish that the mesothelial cells are malignant. Morphologic features that distinguish mesothelioma from reactive conditions include: 1) invasion into adjacent tissue, such as adipose or fibrous tissue, and skeletal muscle; 2) full-thickness serosal involvement; and 3) formation of expansile nodules (considered as a type of fibrous tissue invasion). The presence of tissue invasion is considered to be the most reliable criterion in distinguishing mesothelioma from reactive mesothelial proliferations.[19,20] On the other hand, "worrisome" features such as necrosis, cytologic atypia, and mitoses should be interpreted with caution, since each can be present in reactive pleuritis and do not necessarily indicate malignancy.
- Interpretation can be difficult when there is limited diagnostic tissue, tangential sectioning, artifacts from histologic processing, and/or entrapment of adjacent structures mimicking invasion.[19,21] For a mesothelial proliferation that is suspicious for, but not definitive for malignancy, one may report the findings as "atypical mesothelial proliferation" and recommend rebiopsy and/or close follow-up.
- In the distinction between mesothelioma and benign, reactive mesothelial proliferations, the role of ancillary studies has been limited until recently, where BAP1 or MTAP IHC and *CDKN2A* copy number assessment by FISH may aid in the distinction in some instances (see next page).[22]

References

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PRINCIPLES OF PATHOLOGIC REVIEW

Immunohistochemistry

Markers to Confirm Mesothelial Differentiation

- IHC is integral to the pathologic diagnosis of mesothelioma in clinical practice.
- Useful IHC markers include: 1) positive markers to confirm mesothelial differentiation, such as WT1, calretinin, and D2-40; and 2) negative markers to exclude mimics, such as polyclonal carcinoembryonic antigen (CEA), thyroid transcription factor-1 (TTF-1), and claudin-4.[23-25] One of the caveats is that no individual IHC marker is entirely sensitive and specific. Therefore, it is recommended that a panel including at least two mesothelial markers (eg, calretinin, WT1, D2-40) and two carcinoma markers (eg, claudin-4, TTF-1, polyclonal CEA) should be used to establish the diagnosis.[26]
- Broad-spectrum keratins (eg, AE1/AE3, pancytokeratin, MNF116) are not specific and are expressed in both mesothelioma and carcinomas.
- Sarcomatoid mesothelioma often shows focal to absent expression for most mesothelial markers, with the most sensitive marker being D2-40/podoplanin.[27]
- Recently, GATA3 has been explored as a potential diagnostic marker for sarcomatoid mesotheliomas since GATA3 is expressed in only ~10% to 20% of sarcomatoid carcinoma[28] and strongly expressed in all sarcomatoid/desmoplastic mesotheliomas.[29]

Markers to Confirm a Mesothelial Malignant Proliferation

- Although the distinction between diffuse or localized mesothelioma and reactive mesothelial proliferations primarily relies on histologic assessment, this can be challenging in some cases.
- BAP1, MTAP IHC, and *CDKN2A* (p16) FISH are established markers for diagnosing mesothelioma.[22]
  - BAP1 IHC is a specific (though not sensitive) marker to distinguish mesothelioma from reactive mesothelial proliferations.
  - *BAP1* is a tumor suppressor implicated in the pathogenesis of mesothelioma, uveal melanoma, cholangiocarcinoma, and clear cell renal cell carcinoma.[30] Recurrent somatic and/or germline mutations in *BAP1* are present in mesothelioma. As a surrogate for *BAP1* genomic status, BAP1 IHC is used as a diagnostic marker for mesothelioma, whereas reactive proliferations have intact BAP1 nuclear staining. Complete absence of expression or cytoplasmic staining is considered a loss of BAP1 expression. Aberrant BAP1 protein expression, defined as absence of nuclear BAP1 staining, is present in ~50% to 70% of mesothelioma epithelioid type[31-37] but in <20% of sarcomatoid type.[38]
  - MTAP IHC has been used as a diagnostic marker for mesothelioma.[39] *MTAP* is located near *CDKN2A* on the chromosomal region 9p21. Loss of cytoplasmic MTAP staining is considered a surrogate for chromosomal 9p loss as determined by concurrent *CDKN2A* FISH testing[39] and has been reported in ~40% to 60% of mesothelioma but rarely in reactive proliferations.[35-37]
  - Although MTAP alone is not sensitive, combined use of BAP1 and MTAP IHC may improve sensitivity and specificity.[35-37] Since ~10% to 20% of lung adenocarcinomas have MTAP loss,[36] MTAP IHC is not useful for distinction between mesothelioma and lung carcinoma.
- Additional IHC markers such as 5-hydroxymethylcytosine (5-hmC), enhancer of zeste homolog 2 (EZH2), cyclin D1, and programmed death ligand 1 (PD-L1), and *NF2* by FISH are all potentially useful to distinguish mesothelioma from reactive mesothelial proliferations, but need further study since their utility in clinical practice remains unclear.[22]

References

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PRINCIPLES OF PATHOLOGIC REVIEW

Immunohistochemistry (continued)

Markers as Potential Prognostic and Predictive Markers

- Recent studies explored IHC targets as potential prognostic and predictive markers.
  - Patients with pleural mesothelioma, epithelioid type, with loss of BAP1 by IHC and retained p16 expression by IHC have prolonged survival in both univariate and multivariate analyses.[40]
  - Patients with mesothelioma with germline *BAP1* mutations have a prolonged survival.[41,42]
  - *ALK* rearrangements by IHC found in rare patients with peritoneal mesothelioma[43-46] have shown dramatic response with ALK inhibitor therapies.[47,48]
- PD-L1 (CD274), a negative regulator of immune checkpoint, represents a target in immunotherapy, with PD-L1 IHC evaluated as a predictive biomarker in diverse tumor types.[49]
- The utility of PD-L1 IHC as a predictive marker for immune checkpoint inhibitors and the optimal assessment criteria in mesothelioma remain unclear.

Cytogenetic Features

- Most mesotheliomas are characterized by complex numerical and structural karyotypic alterations.[50]
- Although no specific chromosomal abnormalities are pathognomonic for mesothelioma, loss of chromosomal region 9p including *CDKN2A* or 22q including *NF2* is noted in a subset of tumors.
  - Homozygous loss of *CDKN2A* by FISH testing is present in ~60% of mesotheliomas.[51-53]
  - While detection of *CDKN2A* loss can aid in the distinction of mesothelioma from reactive mesothelial proliferations, *CDKN2A* loss alone is not useful in separating mesothelioma from other tumor types, since *CDKN2A* loss can be found in a substantial fraction of sarcomatoid mesotheliomas, sarcomatoid carcinomas, and sarcomas.[54]
  - Hemizygous loss of *NF2* by FISH is present in ~50% of pleural mesotheliomas.[55]
- A rare subset of pleural mesothelioma harbors a peculiar near-haploid karyotype, with extensive loss of heterozygosity involving nearly all chromosomes except chromosomes 5 and 7.[56]

Molecular Features

- Most mesotheliomas are characterized by recurrent mutations in tumor suppressors and epigenetic regulators, including *BAP1, NF2, TP53, SETD2*, and other genes.[56-60] Consistent with its histomorphologic heterogeneity, mesothelioma shows an impressive molecular diversity.
- Alterations are identified in multiple pathways in the regulation of cell cycle, RNA processing, histone regulation, and cell growth.[58] *BAP1* is one of the most frequently altered genes; mechanisms of *BAP1* inactivation include point mutations, copy number loss, inactivating structural rearrangements, and minute chromosomal deletions.[56-58,61-63]
- Furthermore, a small subset of pleural mesothelioma harbors unusual genetic alterations: Genomic near-haploidization was described in rare pleural mesotheliomas that harbor mutations in *TP53* and/or *SETDB1*.[56]
- Peritoneal mesothelioma has distinct molecular features compared to pleural mesothelioma.[64]

References

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PRINCIPLES OF PATHOLOGIC REVIEW

Molecular Features (continued)

- Oncogenic *EWSR1::ATF1* fusion has been described in pleural and peritoneal mesotheliomas in young adults.[64,65]
  - *ALK* rearrangements have been identified in rare patients with peritoneal mesothelioma.[43-45,48]
- Germline mutations are overall present in 12%–16% of patients with pleural and peritoneal mesotheliomas and primarily involve genes in the DNA repair and cell cycle regulation, such as *BAP1, BRCA2, CDKN2A, TMEM127, VHL, WT1, MRE11A,* and *MSH6*.[42,66,67] Germline mutations appear to be more common in patients who are young, have a family history of mesothelioma, or have a clinical history of other synchronous malignancies.[42,66,68]

Differential Diagnosis

- The differential diagnosis of mesothelioma depends on the histologic type (epithelioid, biphasic, or sarcomatoid) under consideration. Mesothelioma can resemble reactive pleuritis or diverse tumor types, including carcinomas, melanomas, and sarcomas.
- In addition to diffuse mesothelioma, WHO recognizes additional types of mesothelial lesions: 1) localized mesothelioma, 2) WDPMT, and 3) adenomatoid tumor.[1]
  - *Localized pleural mesothelioma* is microscopically identical to mesothelioma, although it is radiographically and grossly solitary and circumscribed.[15-17] Genetically, localized pleural mesothelioma includes three groups (*BAP1*-mutant, *TRAF7*-mutant, and near-haploid), with similarities but also differences from pleural mesothelioma.[18]
  - *WDPMT*, often an incidental finding in the peritoneum of those assigned female at birth, can occur in the pleura,[69] and is genetically characterized by recurrent mutations in *TRAF7* or *CDC42*.[70] Infrequently, WDPMT shows back-to-back papillae with foci of invasion,[71] morphologically mimicking mesothelioma. Furthermore, distinction between a mesothelioma with prominent papillary surface projections and WDPMT can be challenging, particularly in small superficial biopsies.
  - *Adenomatoid tumor* primarily affects the genital tracts but rarely can involve the pleura; recurrent mutations in *TRAF7* have been described in adenomatoid tumors of genital type.[72]
  - *Peritoneal inclusion cyst* is a benign, rare tumor that displays multiple mesothelial-lined cysts that may be distinguished from mesothelial neoplasia. This lesion is almost always located in the peritoneum, although uncommon cases have been described in the pleura. These cystic proliferations are lined by bland mesothelial cells and lack significant stratification, papillary structures, or atypia.
  - *Mesothelioma in situ* is a preinvasive, single-layer surface proliferation of neoplastic mesothelial cells. Since the diagnosis of mesothelioma in situ cannot be simply made on conventional hematoxylin and eosin (H&E) stains, the diagnosis requires either 1) loss of BAP1 nuclear expression by IHC; and/or 2) *CDKN2A* homozygous deletion identified either by FISH or by MTAP IHC (cytoplasmic staining). Furthermore, no mass lesions should be identified on imaging or thoracoscopy.

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PRINCIPLES OF SUPPORTIVE CARE AND SURVIVORSHIP

- **Pleural effusions: Talc pleurodesis or pleural catheter, if required for management of pleural effusion.[a] Drainage is preferred for candidates with potentially operable disease; drainage or pleurodesis are both options for patients with inoperable disease.**
- **Radiotherapy and image-guided thermal ablation[1] are palliative options for symptomatic pleural disease.**
- **Pain management: NCCN Guidelines for Adult Cancer Pain**
- **Nausea/vomiting: NCCN Guidelines for Antiemesis**
- **Psychosocial distress: NCCN Guidelines for Distress Management**
- **Fatigue: NCCN Guidelines for Cancer-Related Fatigue**
- **Immunotherapy-related toxicities: NCCN Guidelines for the Management of Immune Checkpoint Inhibitor-Related Toxicities**
- **Venous thromboembolic disease: NCCN Guidelines for Cancer-Associated Venous Thromboembolic Disease**
- **NCCN Guidelines for Palliative Care as indicated**
- **Smoking cessation counseling and intervention: NCCN Guidelines for Smoking Cessation; NCCN Guidelines for Lung Cancer Screening**
- **Survivorship: NCCN Guidelines for Survivorship**

[a] If FDG-PET/CT is to be done, recommend obtaining FDG-PET/CT before pleurodesis. Confirm diagnosis of pleural mesothelioma prior to pleurodesis. If pleural mesothelioma is suspected, consider evaluation by a multidisciplinary team with expertise in pleural mesothelioma.

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NCCN Guidelines Version 3.2026
Mesothelioma: Pleural

NCCN Guidelines Index
Table of Contents
Discussion

PRINCIPLES OF SYSTEMIC THERAPY[a,b]

HISTOLOGY	FIRST-LINE THERAPY	SUBSEQUENT THERAPY
Epithelioid	**Preferred** • (Carboplatin or Cisplatin)/Pemetrexed[1–4] (category 1) • (Carboplatin or Cisplatin)/Pemetrexed + Bevacizumab[c,5-7] (category 1) • (Carboplatin or Cisplatin)/Pemetrexed + Pembrolizumab[d,8] (category 1) • Ipilimumab + Nivolumab[9] (category 1) **Useful in Certain Circumstances** • (Carboplatin or Cisplatin)/Gemcitabine[10-12] • Pemetrexed[13] • Vinorelbine[14]	Progression → **Preferred** (if chemotherapy first line)[e] • Nivolumab[f,15,16] • Ipilimumab + Nivolumab[15,17] **Preferred** (if Ipilimumab + Nivolumab first line) • (Carboplatin or Cisplatin)/Pemetrexed[1-4] • (Carboplatin or Cisplatin)/Pemetrexed + Bevacizumab[c,5-7] **Other Recommended** • Pemetrexed (category 1)[18,19] • Gemcitabine[20,21] • Gemcitabine + Ramucirumab[22] • Vinorelbine[23,24]
Biphasic or Sarcomatoid	**Preferred** • Ipilimumab + Nivolumab[9] (category 1) • (Carboplatin or Cisplatin)/Pemetrexed + Pembrolizumab[d,8] (category 1) **Other Recommended** • (Carboplatin or Cisplatin)/Pemetrexed[1-4] (category 1) • (Carboplatin or Cisplatin)/Pemetrexed + Bevacizumab[c,5-7] (category 1) **Useful in Certain Circumstances** • (Carboplatin or Cisplatin)/Gemcitabine[10-12] • Pemetrexed[13] • Vinorelbine[14]	Progression → **Preferred** (if Ipilimumab + Nivolumab first line) • (Carboplatin or Cisplatin)/Pemetrexed[1-4] • (Carboplatin or Cisplatin)/Pemetrexed + Bevacizumab[c,5-7] **Preferred** (if chemotherapy first line)[e] • Nivolumab[f,15,16] • Ipilimumab + Nivolumab[15,17] **Other Recommended** • Pemetrexed (category 1)[18,19] • Gemcitabine[20,21] • Gemcitabine + Ramucirumab[22] • Vinorelbine[23,24]

[a] All regimens may also be used for pericardial mesothelioma and tunica vaginalis testis mesothelioma.
[b] Broad molecular tumor profiling is recommended with the goal of identifying rare driver alterations (eg, *NTRK* or *ALK*) for which effective drugs may be available or to appropriately counsel patients regarding the availability of clinical trials.[25]
[c] An FDA-approved biosimilar is an appropriate substitute for bevacizumab.
[d] Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.
[e] Consider rechallenge with pemetrexed-based therapy, if good response to front-line pemetrexed-based treatment.[26]
[f] Nivolumab and hyaluronidase-nvhy is not approved for concurrent use with IV ipilimumab; however, for nivolumab monotherapy, nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.

References on PM-C (2 of 3)

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PRINCIPLES OF RADIATION THERAPY

General Principles

- Recommendations regarding RT should be made by radiation oncologists with experience in managing pleural mesothelioma.
- The best timing for delivering RT after surgical intervention and/or in conjunction with chemotherapy should be discussed by a multidisciplinary team including radiation oncologists, surgeons, medical oncologists, diagnostic imaging specialists, and pulmonologists.
- PET scanning for treatment planning can be used as indicated.
- Prophylactic RT is not routinely recommended to prevent instrument-tract recurrence after pleural intervention.[1]
- RT is an effective palliative treatment for relief of chest pain, bronchial or esophageal obstruction, or other symptomatic sites associated with mesothelioma.
- A randomized phase III trial in patients with non-metastatic pleural mesothelioma who underwent non-radical lung-sparing surgery found substantially greater overall survival with sequential pleural intensity-modulated RT (IMRT) compared to palliative RT.[2] Sequential pleural IMRT after P/D in the presence of an intact lung may be considered in centers with experience and expertise in these methods, given the technical difficulty of this treatment.[3-5]
- Advanced technologies may be used, such as image-guided RT (IGRT) for treatment involving IMRT/stereotactic radiosurgery (SRS)/ stereotactic body RT (SBRT), and intensity-modulated proton therapy (IMPT).[6]

Radiation Dose and Volume

- The dose of radiation should be based on the purpose of the treatment. See Recommended Doses for Radiation Therapy (PM-D 2 of 3).
- A dose ≥60 Gy should be delivered to macroscopic residual tumors if the doses to adjacent normal structures are limited to their tolerances. In addition to covering the surgical bed within the thorax, the volume of postoperative radiation should also include the surgical scars and biopsy tracks in the chest wall.[7-9]
- Daily doses of 4 Gy appear to be more efficacious than fractions of <4 Gy in providing relief from chest pain associated with mesothelioma,[8,10] although the optimal daily and total dose of RT for palliative purposes remains unclear.
- For patients with residual tumors, some experienced investigators have used brachytherapy or intraoperative external beam RT (EBRT) in combination with surgery.

Radiation Techniques (PM-D 2 of 3)
References (PM-D 3 of 3)

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PRINCIPLES OF RADIATION THERAPY

Recommended Doses for Radiation Therapy

Treatment type	Total dose	Fraction size	Treatment duration
Palliative			
Chest wall pain from recurrent nodules	20–40 Gy or 30 Gy	≥4 Gy 3 Gy	1–2 weeks 2 weeks
Multiple brain or bone metastases	30 Gy	3 Gy	2 weeks
Post P/D Higher dose to higher risk areas	45–60 Gy	1.8–2 Gy	5–6 weeks

Radiation Techniques

- Use of highly conformal radiation technology (IMRT) is the preferred choice based on comprehensive consideration of target coverage and clinically relevant normal tissue tolerance.[4,11] Advanced technologies are appropriate when needed to deliver curative RT safely. These technologies include (but are not limited to) 4D-CT and/or PET/CT simulation, IMRT/volumetric modulated arc therapy (VMAT), IGRT, motion management, and proton therapy.
- Special attention should be paid to minimize radiation to the contralateral lung,[12] as the risk of fatal pneumonitis with IMRT is excessively high when strict limits are not applied.[13] The contralateral uninvolved mean lung dose (MLD) should be kept as low as possible, preferably <8.5 Gy. The low-dose volume should be minimized.[14] For postoperative RT for patients who have P/D, other recommended specific lung-preserving techniques are advised. Limit the ipsilateral lung dose to decrease risk of pneumonitis and keep total MLD <21 Gy and V20 <40% and contralateral lung V20 <7% and MLD <8 Gy.[15]
- The gross tumor volume (GTV) should include any grossly visible tumor. Surgical clips (indicative of gross residual tumor) should be included for postoperative adjuvant RT.
- The clinical target volume (CTV) for adjuvant RT after P/D should encompass the entire pleural surface (for partial resection cases), surgical clips, and any potential sites with residual disease.
- Extensive elective nodal irradiation (ENI) (entire mediastinum and bilateral supraclavicular nodal regions) is not recommended.
- The planning target volume (PTV) should consider the target motion and daily setup errors. The PTV margin should be based on the individual patient's motion, simulation techniques used (with and without inclusion motion), and reproducibility of each clinic's daily setup.

General Principles and Radiation Dose and Volume (PM-D 1 of 3)

References (PM-D 3 of 3)

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PRINCIPLES OF SURGERY[1]

- **Surgical resection should be performed on carefully evaluated patients by thoracic surgeons with experience in managing pleural mesothelioma.**
- **Decisions regarding surgical options for treatment are highly dependent on accurate histology. Pleural biopsy for diagnosis should provide enough tissue for differentiation of epithelioid, sarcomatoid, or mixed histology and clearly exclude metastatic pleural involvement of another primary. Cytology is generally not considered adequate for important histologic differentiation required for treatment decisions.**
- **For patients being considered for surgery, a single-port thoracoscopy on the line of the potential incision is recommended.**
- **The goal of surgery is complete gross cytoreduction of the tumor. The goal of cytoreductive surgery is "macroscopic complete resection"—in other words, removal of ALL visible or palpable tumors. In cases where this is not possible, such as in multiple sites of chest wall invasion, surgery should be aborted. If it is possible to remove most of the gross disease to help with postoperative management, with a minimal impact on morbidity, then surgery should be continued.**
- **The surgical choices are: 1) P/D with mediastinal lymph node sampling, which is defined as complete removal of the pleura and all gross tumor ± en-bloc resection of pericardium and/or diaphragm with reconstruction; and 2) EPP, which is defined as en-bloc resection of the pleura, lung, ipsilateral diaphragm, and often pericardium. Mediastinal node sampling should be performed with a goal to obtain at least 3 nodal stations.**
- **For early-stage disease (confined to the pleural envelope, no clinical evidence for lymph node involvement) with favorable histology (epithelioid), P/D may be considered but it is unclear whether there is benefit over systemic therapy alone. There is controversy regarding a decision for surgical resection that needs to be weighed, taking into account tumor histology, distribution, the patient's pulmonary reserve, and availability of adjuvant and intraoperative strategies. P/D is the preferred surgical treatment option and can be considered in select patients for complete gross cytoreduction. EPP may be selected in certain cases that require careful consideration of the total treatment plan that includes the patient and multidisciplinary team.[2-5]**
- **If N2 disease is identified, prognosis with surgery (and other therapy) is substantially diminished. Surgical resection should only be considered in the setting of a clinical trial or at a center with expertise in pleural mesothelioma.**
- **If technically appropriate for even more advanced disease, lung-sparing operations like P/D reduce the risk for perioperative mortality and may be acceptable in terms of achieving complete macroscopic resection. P/D can provide excellent symptomatic control of recurrent pleural effusions.**
- **Intraoperative adjuvant therapy is still under investigation but may be considered as part of a reasonable multidisciplinary approach to this locally aggressive disease.**
- **After recovery from surgery, patients should be referred for adjuvant therapy, which may include chemotherapy and RT depending on whether any preoperative therapy was used and on the pathologic analysis of the surgical specimen.**

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Table 1. **Definitions for T, N, M (9th Edition)**

Clinical T (cT)

T	T Criteria
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
T1	Tumor limited to the ipsilateral pleura Psum† ≤12 mm and no involvement of the fissure (Fmax‡ ≤5 mm)
T2	Tumor involving each of the ipsilateral pleura with Psum† ≤12 mm, and with any of the following: - involvement of the fissure (Fmax‡ >5 mm) - mediastinal fat invasion - solitary area of chest wall soft tissue invasion OR Tumor involving each of the ipsilateral pleura with Psum† >12 mm but ≤30 mm, with or without: - involvement of the fissure (Fmax‡ >5 mm) - mediastinal fat invasion - solitary area of chest wall soft tissue invasion
T3	Tumor involving the ipsilateral pleura with Psum† >30 mm, with or without: - involvement of the fissure (Fmax‡ >5 mm) - mediastinal fat invasion - solitary area of chest wall soft tissue invasion
T4	Tumor with invasion of any of the following: - chest wall bony invasion (rib) - mediastinal organs (heart, spine, esophagus, trachea, great vessels) - diffuse chest wall invasion - direct tumor extension through the diaphragm or pericardium - direct tumor extension to the contralateral pleura - presence of malignant pericardial effusion

Pathological T (pT)

T	T Criteria
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
T1	Tumor limited to the ipsilateral pleura with no involvement of the fissure
T2	Tumor involving the ipsilateral pleura with any of the following: - involvement of the fissure - ipsilateral lung parenchyma invasion - diaphragm (non-transmural) invasion
T3	Tumor limited to the ipsilateral pleura (with or without fissure involvement) and with invasion of any of the following: - mediastinal fat - surface of the pericardium - endothoracic fascia - solitary area of chest wall soft tissue
T4	Tumor with invasion of any of the following: - chest wall bony invasion (rib) - mediastinal organs (heart, spine, esophagus, trachea, great vessels) - diffuse chest wall invasion - transmural invasion of the diaphragm or pericardium - direct extension to the contralateral pleura - presence of malignant pericardial effusion

† Psum = pmax1+pmax2+pmax3 (sum of 3 measurements of maximal pleural thickness measured on axial images along the chest wall or mediastinum in each of the three divisions of the chest -- upper, middle, and lower divided by two lines: one at the top of the aortic arch and the second drawn at the top of the left atrium)
‡ Fmax = maximal thickness of pleural tumor along the fissures measured on the sagittal images

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Table 1. **Definitions for T, N, M**

N	**Regional Lymph Nodes**
NX	Regional lymph nodes cannot be assessed
N0	No tumor involvement of regional lymph node(s)
N1	Tumor involvement of ipsilateral bronchopulmonary, hilar, or mediastinal (including the internal mammary, peridiaphragmatic, pericardial fat pad, or intercostal lymph nodes) regional lymph nodes
N2	Tumor involvement of contralateral mediastinal, ipsilateral, or contralateral supraclavicular lymph nodes
M	**Distant Metastasis**
cM0	No distant metastasis
cM1	Distant metastasis
pM1	Microscopic confirmation of distant metastasis

Table 2. **AJCC Prognostic Groups (9th Edition)**

	T	N	M
Stage I	T1	N0	M0
Stage II	T2	N0	M0
	T1	N1	M0
Stage IIIA	T2	N1	M0
	T3	N0-N1	M0
	T1-T3	N2	M0
Stage IIIB	T4	Any N	M0
Stage IV	Any T	Any N	M1

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ABBREVIATIONS

3D-CRT	three-dimensional conformal radiation therapy
4D-CT	four-dimensional computed tomography
5-hmC	5-hydroxymethylcytosine
CEA	carcinoembryonic antigen
CTV	clinical target volume
DLCO	diffusing capacity of the lung for carbon monoxide
EBRT	external beam radiation therapy
EBUS/ EUS	endobronchial ultrasound/ endoscopic ultrasound
ENI	elective nodal irradiation
EPP	extrapleural pneumonectomy
EZH2	enhancer of zeste homolog 2
FDG	fluorodeoxyglucose
FISH	fluorescence in situ hybridization
FNA	fine-needle aspiration
GTV	gross tumor volume
H&E	hematoxylin and eosin
IGRT	image-guided radiation therapy
IHC	immunohistochemistry
IMPT	intensity-modulated proton therapy
IMRT	intensity-modulated radiation therapy
MLD	mean lung dose
NGS	next-generation sequencing
P/D	pleurectomy/decortication
PD-L1	programmed death ligand 1
PFT	pulmonary function test
PS	performance status
PTV	planning target volume
SBRT	stereotactic body radiation therapy
SMRP	soluble mesothelin-related peptide
SRS	stereotactic radiosurgery
TTF-1	thyroid transcription factor-1
VATS	video-assisted thoracic surgery
VMAT	volumetric modulated arc therapy
WDPMT	well-differentiated papillary mesothelial tumor

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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NCCN Guidelines Version 3.2026
Mesothelioma: Pleural

Discussion

This discussion corresponds to the NCCN Guidelines for Mesothelioma: Pleural. Last updated: July 8, 2026

Table of Contents

Overview MS-2

Guidelines Update Methodology MS-2

Literature Search Criteria MS-2

Sensitive/Inclusive Language Usage MS-2

Risk Factors MS-3

Diagnosis MS-3

Pretreatment Evaluation MS-5

Overview of Treatment Approaches MS-5

- Systemic Therapy MS-5
- Surgery MS-9
- Radiation Therapy MS-11

Treatment Recommendations for Pleural Mesothelioma MS-12

- Clinical Stage I PM with Epithelioid Histology MS-12
- Clinical Stage II-IV PM with Epithelioid Histology; Sarcomatoid or Biphasic Histology (Any Stage); or Medically Inoperable MS-13

Summary MS-14

References MS-15

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Overview

Mesothelioma is a rare cancer originating from mesothelial surfaces of the pleura and other sites that is estimated to occur in approximately 3000 people in the United States every year.[1,2] Although the incidence has stabilized in the United States in recent years, mesothelioma diagnoses continue to increase worldwide.[3] Pleural mesothelioma (PM), historically referred to as malignant pleural mesothelioma, is the most common type (approximately 85%); however, mesothelioma can also occur in the lining of other sites, such as the peritoneum (approximately 15%), pericardium, and tunica vaginalis testis.[2,4,5] PM is difficult to treat, as most patients have advanced disease at presentation.[6,7] Median overall survival (OS) is approximately 1 year after diagnosis of PM, and 5-year OS is about 5% to 10%.[7-14] Most patients diagnosed with PM are >60 years of age.[6,9] PM occurs mainly in males who have been exposed to asbestos, although diagnosis may occur decades after exposure.[6,7,15,16] For example, it has been estimated that the median latent period is >32 years.[15]

These NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) provide recommendations for the evaluation and treatment of patients diagnosed with PM. The Guidelines were developed and are updated by Panel members who also update the NCCN Guidelines for Mesothelioma: Peritoneal and the NCCN Guidelines for Non-Small Cell Lung Cancer.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria

Prior to the annual update of the NCCN Guidelines for Mesothelioma: Pleural, an electronic search of the PubMed database was performed to obtain key literature in pleural mesothelioma published since the previous Guidelines update, using the search term: pleural mesothelioma. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[17]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase II; Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Practice Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these Guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[18] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender

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individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Risk Factors

Exposure to asbestos is the primary risk factor for PM.[5,19,20] Most mesotheliomas are associated with occupational exposure to asbestos; however, individuals with non-occupational exposure also have increased risk.[21-23] Some patients exposed to asbestos may only have benign pleural disease.[24,25]

Other risk factors for PM have been identified. Patients exposed to ionizing radiation are thought to be at increased risk of developing mesothelioma.[20,26-37] One study that evaluated 47,219 patients with Hodgkin lymphoma and 252,090 patients with non-Hodgkin lymphoma from the Surveillance, Epidemiology, and End Results (SEER) database found that patients treated with radiation therapy (RT) had a higher risk of developing mesothelioma as a secondary malignancy than those who were not irradiated.[26] Another risk factor is erionite, a naturally occurring fiber-like mineral that is more potent than asbestos; however, fewer people are at risk of developing mesothelioma from erionite as it is only found in certain geographic locations and is generally not used for commercial applications.[38-42]

Genetic factors may also play a role in PM.[38,43-52] For example, a high incidence of PM has been reported in a small number of families in the absence of exposure to occupational asbestos or erionite; germline mutations in the *BAP1* gene were identified as a risk factor.[52] Interestingly, some studies reported that patients with PM and deleterious germline *BAP1* mutations have prolonged survival compared to those with wild-type *BAP1*.[47,50] Genetic counseling and germline testing may be warranted for patients with a young age at diagnosis, personal or family history suggestive of *BAP1* tumor predisposition syndrome, or multi-*BAP1*-associated malignancies.

Smoking is not considered a risk factor for mesothelioma.[53,54] However, patients who smoke and have been exposed to asbestos are at increased risk for lung cancer.[55,56] Patients who smoke should be encouraged to quit because it may negatively affect treatment.[57] Please refer to the NCCN Guidelines for Smoking Cessation, available at www.NCCN.org.

Although screening for mesothelioma has been studied in patients at high risk (ie, those with asbestos exposure), the NCCN Guidelines do not recommend screening for PM because it has not been shown to decrease mortality.[55,58-64] Note that the NCCN Guidelines for Lung Cancer Screening with low-dose CT do not apply to PM as there are no data to suggest that screening with low-dose CT improves survival for patients with PM.[55,61,65-68]

Diagnosis

Presentation and Evaluation

Patients with suspected PM often have dyspnea and chest wall pain; they may also have pleural effusion, fatigue, insomnia, cough, chest wall mass, loss of appetite, and weight loss.[5,54,69,70] Patients with PM may have a high symptom burden when compared with patients who have other types of cancer. Most patients present without distant metastases, and central nervous system (CNS) metastases are rare.[71]

In patients with recurrent pleural effusion and/or pleural thickening, the recommended initial evaluation for suspected PM includes: 1) chest CT with contrast; 2) thoracentesis for cytologic assessment; and 3) pleural biopsy (eg, thoracoscopic biopsy [preferred], CT-guided core biopsy, open biopsy) (see PM-1 in the algorithm).[69,72-79] However, cytologic samples are often negative even when patients have PM.[80,81] Soluble

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mesothelin-related peptide (SMRP) levels may also be assessed, and these levels may correlate with disease status.[82-85]

Pleurodesis or pleural catheter may be needed for management of pleural effusion (see *Principles of Supportive Care and Survivorship* in the algorithm).[86-94] Drainage can be pursued in patients who are awaiting systemic therapy, including use of indwelling tunneled pleural catheters.[86] Patients should be assessed for distress. Refer to the NCCN Distress Thermometer and Problem List within the NCCN Guidelines for Distress Management (DIS-A) at www.NCCN.org.

Pathology

The goals of the pathologic evaluation of mesothelioma are to confirm the pathologic diagnosis and to determine histology. The diagnostic tools used include histologic assessment, immunohistochemistry (IHC), cytogenetics, and molecular techniques (eg, targeted next-generation sequencing [NGS], fluorescence in situ hybridization [FISH], single-nucleotide polymorphism arrays). However, the diagnosis relies primarily on proper histologic assessment and IHC.

There are three histologic subtypes of mesothelioma according to the World Health Organization (WHO) histologic classification system based on cytologic features: epithelioid, sarcomatoid, and biphasic (mixed with both epithelioid and sarcomatoid components, with each comprising ≥10% of the tumor).[95] Epithelioid histology is the most common type and has a more favorable prognosis than biphasic or sarcomatoid histologies.[9,96] Assignment of histologic type can be challenging given the intertumoral and intratumoral morphologic heterogeneity; however, it is essential to determine the histology for prognostication and treatment planning purposes.

It is sometimes difficult to distinguish localized mesothelioma from diffuse mesothelioma.[97] Almost all mesotheliomas are diffuse (>99%); however, rare cases of localized mesothelioma have been diagnosed, which are less aggressive.[97-101] Distinguishing malignant disease from benign pleural disease (such as reactive pleuritis) and from other types of malignancies, such as carcinomas, melanomas, and sarcomas, can also be challenging.[102-105]

IHC can be used to confirm mesothelial differentiation and mesothelial malignant proliferation. Cytologic samples of pleural fluid are often negative or inconclusive, but diagnosis can sometimes be made using cytology.[80,81,106,107] IHC markers used to confirm mesothelioma differentiation include markers specific for mesothelial differentiation (eg, WT1, calretinin, D2-40) and markers that typically are positive in carcinoma and negative in mesothelioma (eg, thyroid transcription factor 1 [TTF-1], polyclonal carcinoembryonic antigen [CEA], claudin-4) (see *Principles of Pathologic Review* in the algorithm).[4,108-111] A panel of two positive mesothelial markers and two negative carcinoma markers is recommended for diagnosis of mesothelioma.[4] The presence or absence of *BAP1* nuclear expression assessed by IHC can also be used in the diagnosis of mesothelioma.[4,112,113] MTAP IHC and *CDKN2A* (p16) FISH are other established markers that can be used to inform a mesothelioma diagnosis.[4] Other markers may also be useful in establishing mesothelial malignant proliferation, but additional data are needed.[114]

Most mesotheliomas are characterized by complex karyotypic alterations as well as by recurrent mutations in tumor suppressors and epigenetic regulators. Rare genetic alterations such as *EWSR1-ATF1* fusions, *TP53, NF2*, *SETDB1,* or *SETD2* mutations have been found in patients with PM.[115-120] *NTRK* and *ALK* gene fusions have also been identified.[121-123] PMs generally have distinct molecular features compared with peritoneal mesotheliomas.[124]

See the *Principles of Pathologic Review* section in the algorithm for additional information on pathologic evaluation.

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Pretreatment Evaluation

Once a PM diagnosis is confirmed, patients should be treated by a multidisciplinary team with experience in PM (see PM-1 in the algorithm). The multidisciplinary team should include diagnostic imaging specialists, pulmonologists, surgeons, medical oncologists, and radiation oncologists. A chest and abdominal CT with contrast is recommended as part of the pretreatment evaluation for staging and to guide treatment strategy for patients (see PM-2 in the algorithm).[73,74,79,125]

Patients with a diagnosis of clinical stage I PM with epithelioid histology may undergo additional testing (see PM-3 in the algorithm), including 1) fluorodeoxyglucose (FDG)-PET/CT; 2) mediastinoscopy or endobronchial ultrasound (EBUS)/endoscopic ultrasound (EUS) fine-needle aspiration (FNA) of the mediastinal lymph nodes[126,127]; and 3) optional chest MRI with contrast to evaluate possible chest wall, spinal, diaphragmatic, or vascular involvement.[73,79] Video-assisted thoracoscopic surgery (VATS) or laparoscopy can be considered if contralateral or peritoneal disease is suspected to rule out transdiaphragmatic extension (eg, extension to the peritoneum is indicative of stage IV [unresectable] disease).[128]

FDG-PET/CT scans should be obtained before pleurodesis if practical, because talc produces pleural inflammation, which can affect FDG avidity (ie, false-positive result),[129-131] or after drainage of effusion without pleurodesis to better evaluate pleural disease burden.

Patients with clinical stage I PM with epithelioid histology should be evaluated to assess whether they can tolerate surgery using pulmonary function tests (PFTs), including diffusing capacity of the lung for carbon dioxide (DLCO) and cardiac stress tests.

Overview of Treatment Approaches

Systemic Therapy

The current Guidelines recommend systemic therapy as the primary treatment modality for most patients with PM, regardless of resectability (see PM-2 in the algorithm). For the minority of patients who will receive cytoreductive surgery, systemic therapy is recommended as either induction or adjuvant therapy.

First-Line Therapy

Historically, cytotoxic chemotherapy has been the primary type of systemic therapy used for the treatment of PM. Pemetrexed in combination with cisplatin is approved by the U.S. Food and Drug Administration (FDA) as an initial treatment for patients with PM whose disease is unresectable or who are otherwise not candidates for curative surgery.[132] This approval was based on data from a phase 3 randomized trial that assessed cisplatin/pemetrexed versus cisplatin alone in 456 patients with PM who were not candidates for surgery.[133] A longer median survival time was observed in the cisplatin/pemetrexed group compared with cisplatin alone (12.1 vs. 9.3 months, $P = .02$).

Carboplatin/pemetrexed was assessed in three separate phase 2 studies; the reported median OS ranged from 12.7 to 14 months.[134-136] An expanded access trial that evaluated cisplatin/pemetrexed or carboplatin/pemetrexed in 1704 chemotherapy-naïve patients with medically inoperable PM demonstrated that the two regimens resulted in similar 1-year survival rates.[137]

A multicenter phase 3 randomized trial (IFCT-GFPC-0701 MAPS) evaluated the addition of bevacizumab to cisplatin/pemetrexed (with maintenance bevacizumab) versus cisplatin/pemetrexed alone for 448 patients with unresectable PM and performance status (PS) 0 to 2 who were chemotherapy-naive and did not have any substantial cardiovascular

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comorbidities.[138] The median OS was longer in the bevacizumab plus cisplatin/pemetrexed arm compared with cisplatin/pemetrexed alone (18.8 vs. 16.1 months; hazard ratio [HR], 0.77; *P* = .0167). Grade 3 to 4 adverse events (AEs) were reported in 71% (158/222) of patients receiving the bevacizumab-containing regimen compared with 62% (139/224) of those receiving cisplatin/pemetrexed alone. More grade 3 or higher hypertension (23% vs. 0%), grade 3 proteinuria (3% vs. 0%), and grade 3 to 4 thrombotic events (6% vs. 1%) were observed in patients in the bevacizumab plus cisplatin/pemetrexed arm than in those in the cisplatin/pemetrexed group.

A phase 2 open-label study assessed bevacizumab in combination with carboplatin/pemetrexed as a first-line therapy in 76 patients with unresectable PM.[139] The median OS was 15.3 months. A partial response was observed in 34.2% (26/76) of patients, while 57.9% (44/76) had stable disease. Bevacizumab is not recommended for those with uncontrolled hypertension, risk for bleeding or clotting, and substantial cardiovascular morbidity.[79]

Clinical benefit with other chemotherapy agents has also been reported in the first-line setting. Cisplatin or carboplatin in combination with gemcitabine may be useful for patients who cannot receive pemetrexed; partial responses with these regimens have been reported in phase 2 studies.[140-143] Other first-line options include pemetrexed or vinorelbine for patients who are not candidates for platinum-based combination therapy.[144-146]

More recently, two immune checkpoint inhibitor (ICI)-containing regimens have demonstrated efficacy in the treatment of PM in the first-line setting. Nivolumab in combination with ipilimumab is approved by the FDA for the treatment of adult patients with unresectable PM as a first-line treatment.[132] This approval was based on data from CheckMate 743, a phase 3 open-label randomized trial that assessed nivolumab plus ipilimumab versus platinum/pemetrexed chemotherapy as first-line therapy in 605 patients with unresectable PM.[147-149] The majority of patients had epithelioid histology (75%). Based on the 5-year follow-up, the median OS was 18.1 months in patients receiving nivolumab plus ipilimumab versus 14.1 months in those receiving chemotherapy (HR, 0.75).[149]

The trial was not powered to assess superiority within the subgroups; however, among 455 patients with epithelioid histology, the median OS was 18.2 months in those receiving nivolumab plus ipilimumab versus 16.7 months in those receiving chemotherapy (HR, 0.85 [95% CI, 0.69 to 1.03]).[149] Among 150 patients with non-epithelioid histology, the median OS was 18.1 months in patients receiving nivolumab plus ipilimumab versus 8.8 months in those receiving chemotherapy (HR, 0.48 [95% CI, 0.33 to 0.68]).[149] The median OS among those treated with nivolumab plus ipilimumab was similar between the programmed death ligand 1 (PD-L1) <1% and PD-L1 ≥1% groups.[149] Grade 3 to 4 treatment-related AEs were similar in both groups: 31% of patients receiving nivolumab plus ipilimumab and 32% of those receiving chemotherapy.[148]

Pembrolizumab in combination with platinum and pemetrexed chemotherapy is also FDA approved as a first-line treatment for adult patients with unresectable advanced or metastatic PM.[132] The efficacy and safety of pembrolizumab in combination with chemotherapy (pemetrexed with cisplatin or carboplatin) versus chemotherapy alone in the first-line setting was evaluated in 440 patients with advanced PM in a randomized phase 3 trial.[150] Epithelioid histology represented over 75% of the study population. The median OS was significantly longer with pembrolizumab and chemotherapy than with chemotherapy alone (17.3 vs. 16.1 months; HR, 0.79; *P* = .0324). Subgroup data indicate that there was a survival benefit with pembrolizumab and chemotherapy regardless of PD-L1 levels.

Among 95 patients in the non-epithelioid subgroup, the median OS was 12.3 months in the pembrolizumab plus chemotherapy arm and 8.2

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months in the chemotherapy only arm (HR, 0.57 [95% CI, 0.36 to 0.89]).[150] Among 345 patients with epithelioid histology, the median OS was 19.8 months in the pembrolizumab plus chemotherapy group compared with 18.2 months in the chemotherapy only group (HR, 0.89 [95% CI, 0.70 to 1.13]).[150] Grade 3 or 4 treatment-related AEs were reported in 27% of the pembrolizumab plus chemotherapy group and 15% of the chemotherapy only group.[150]

Data from both trials evaluating first-line ICI regimens showed that immunotherapy led to similar OS outcomes regardless of histology. In contrast, non-epithelioid PM appeared to be less chemosensitive than epithelioid PM; chemotherapy only regimens resulted in worse survival outcomes in the non-epithelioid subgroup than the epithelioid subgroup in both trials. Therefore, ICI-based therapy is favored for PM with non-epithelioid histology.

Not all chemoimmunotherapy regimens confer a clear survival benefit over standard chemotherapy. In the randomized phase 3 BEAT-meso trial, 400 patients with advanced PM were randomized to receive atezolizumab plus carboplatin/pemetrexed and bevacizumab or carboplatin/pemetrexed and bevacizumab alone as first-line therapy.[151] The median OS was 20.5 months in the atezolizumab plus carboplatin/pemetrexed and bevacizumab arm and 18.1 months in the carboplatin/pemetrexed and bevacizumab arm; however, this difference in OS was not statistically significant (HR, 0.84; *P* = .14). Given these data, atezolizumab in combination with chemotherapy is not included in the Guidelines as a treatment option.

NCCN Recommendations

Two ICI-containing regimens are recommended as Preferred and category 1 first-line systemic therapy options for eligible patients with PM, regardless of histology: ipilimumab in combination with nivolumab and pembrolizumab in combination with (carboplatin or cisplatin)/pemetrexed. Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for intravenous (IV) pembrolizumab. PD-L1 testing is not required for prescribing an ICI-containing regimen for patients with PM. Immune-related AEs, such as pneumonitis, may occur with ICI treatment (refer to the NCCN Guidelines for Management of Immune Checkpoint Inhibitor-Related Toxicities, available at www.NCCN.org).[152-154]

The NCCN Panel also recommends (carboplatin or cisplatin)/pemetrexed with or without bevacizumab as category 1 first-line therapy options for patients with PM. These are included in the NCCN Guidelines as Preferred options for PM with epithelioid histology, and as Other Recommended options for PM with biphasic or sarcomatoid histology. An FDA-approved biosimilar is an appropriate substitute for bevacizumab. The Panel recommends the following as regimens that may be Useful in Certain Circumstances, regardless of histology: 1) (carboplatin or cisplatin)/gemcitabine; 2) pemetrexed; or 3) vinorelbine.

All regimens recommended for PM can also be used for pericardial mesothelioma and tunica vaginalis testis mesothelioma; however, these recommendations are based on clinical evidence extrapolated from PM due to the limited availability of prospective data in these rare mesotheliomas.[155,156] Broad molecular tumor profiling is recommended by the NCCN Panel with the goal of identifying rare driver alterations (eg, *NTRK* or *ALK*) for which effective drugs may be available or to appropriately counsel patients regarding the availability of clinical trials.[121] Refer to the *Principles of Systemic Therapy* (PM-C 1 of 3) within the algorithm for a complete list of recommended systemic therapy options for PM.

Subsequent Therapy

The approach to subsequent therapy in patients with PM has generally not been well established due to limited data from randomized clinical trials;

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however, there are several viable options depending on the type of systemic therapy given in the first-line setting.

The clinical benefit of nivolumab (with or without ipilimumab) as subsequent systemic therapy for patients with PM who have not received prior immunotherapy was demonstrated in several trials.[157-159] The phase 3 randomized controlled CONFIRM trial assessed nivolumab (67%) versus placebo (33%) in 332 patients with mesothelioma after progression on platinum-based chemotherapy.[157] Most patients had PM (95%) and epithelioid histology (88%). Median OS was 10.2 months in the nivolumab group versus 6.9 months in those receiving placebo (HR, 0.69; *P* = .009). A clinical benefit was observed regardless of PD-L1 expression status. Diarrhea and infusion-related reactions were the most common grade 3 or higher AEs reported with nivolumab. Serious AEs were similar between the groups (41% for nivolumab vs. 44% for placebo).

A phase 2 randomized trial (IFCT-1501 MAPS2) assessed nivolumab with or without ipilimumab as subsequent therapy for 125 patients with PM who previously received platinum-based therapy in combination with pemetrexed.[158] Within the intention-to-treat population, the 12-week disease control rate (DCR) was 40% in the nivolumab/ipilimumab group and 52% in the nivolumab only group. The median OS reported in this non-comparative trial was 15.9 months in the nivolumab/ipilimumab arm and 11.9 months with nivolumab alone.[158] High PD-L1 expression ≥25% was associated with greater overall response based on an exploratory analysis of trial data; however, only a few patients had ≥50% PD-L1 levels. There were more grade 3 to 4 AEs with nivolumab/ipilimumab than with nivolumab alone (26% vs. 14%).[158]

For patients who received first-line nivolumab in combination with ipilimumab, the optimal treatment following disease progression has not been determined in randomized clinical trials. However, given the strength of data supporting the efficacy of platinum-based chemotherapy regimens in the first-line setting, carboplatin or cisplatin in combination with pemetrexed (with or without bevacizumab) are appropriate options in the subsequent-line setting if not given previously.[133,135-139,151]

There is clinical evidence to support the use of other chemotherapy-based regimens for the treatment of PM in the subsequent-line setting. One phase 3 randomized trial assessed pemetrexed plus best supportive care as subsequent therapy versus best supportive care alone in 243 patients with PM after progression on chemotherapy (excluding pemetrexed).[160] The difference in median OS was not statistically significant between the arms (8.4 months for pemetrexed and supportive care vs. 9.7 months for supportive care only; *P* = .74); however, this result may be explained in part due to the design of the study, as patients could cross over to pemetrexed.

Single-agent vinorelbine and single-agent gemcitabine may also be given as subsequent therapy for patients with PM.[161-165] PROMISE-meso, a multicenter phase 3 randomized trial, assessed pembrolizumab versus either gemcitabine or vinorelbine as subsequent therapy in 144 patients with PM following disease progression on platinum-based chemotherapy.[165] There was no significant difference in OS between the groups (HR, 1.12; *P* = .59). Based on the data from this trial and other published studies, the Panel continues to support the inclusion of single-agent vinorelbine and single-agent gemcitabine rather than single-agent pembrolizumab as subsequent therapy options.

The phase 2 randomized RAMES trial evaluated gemcitabine with placebo versus gemcitabine with ramucirumab as subsequent therapy in 161 patients with PM after progression on platinum/pemetrexed chemotherapy.[162] The median OS was significantly longer in patients who received both gemcitabine and ramucirumab than in those who received gemcitabine and placebo (13.8 vs. 7.5 months; HR, 0.71; *P* = .028). Grade 3 or 4 treatment-related AEs occurred in 44% of the patients in the

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gemcitabine and ramucirumab group and 30% of the gemcitabine and placebo group.

For patients who received pemetrexed as part of first-line therapy, re-challenge may be effective for certain patients, particularly if there was a good disease response following first-line pemetrexed.[166,167] A retrospective multicenter database study found that rechallenging with a pemetrexed/platinum regimen reduced the risk of death when compared with rechallenging with pemetrexed alone (HR, 0.11; $P < .001$).[167]

NCCN Recommendations
If chemotherapy (and no immunotherapy) was administered as first-line therapy, the NCCN Panel recommends nivolumab with or without ipilimumab as Preferred subsequent therapy options, regardless of histology. For nivolumab monotherapy, nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for the IV formulation.

Rechallenge with pemetrexed-based therapy can also be considered if there was a good response to front-line pemetrexed-based therapy.

If ipilimumab/nivolumab combination therapy was administered as first-line therapy, the Panel recommends (carboplatin or cisplatin)/pemetrexed (with or without bevacizumab) as Preferred subsequent therapy options.

The other subsequent therapy options recommended by the NCCN Panel are: 1) pemetrexed (category 1); 2) vinorelbine; or 3) gemcitabine (with or without ramucirumab).

See *Principles of Systemic Therapy* (PM-C 1 of 3) within the algorithm for a complete list of recommended systemic therapy options for PM.

Surgery

For eligible patients with PM who undergo surgical evaluation and are deemed candidates, the goal of cytoreductive surgery is to achieve macroscopic complete resection by removing all visible or palpable tumors.[168] If macroscopic complete resection is not possible, such as in the presence of multiple sites of chest wall invasion, then surgery should be aborted. However, surgery should be continued if most of the gross disease can be removed to help with postoperative management and if there will be minimal impact on morbidity.

Two types of cytoreductive surgical procedures are generally used to treat patients with PM: pleurectomy/decortication (P/D) and extrapleural pneumonectomy (EPP).[168] P/D, also known as total pleurectomy, is a lung-sparing approach that involves the complete removal of the involved pleura and all gross tumor with or without en-bloc resection of the pericardium and/or diaphragm. Extended P/D consists of resection of the diaphragm and pericardium in addition to total pleurectomy. EPP refers to en-bloc resection of the involved pleura, lung, ipsilateral diaphragm, and often the pericardium (see *Principles of Surgery* in the algorithm). Mediastinal nodal dissection should be performed with a goal of obtaining at least three nodal stations (see the NCCN Guidelines for Non-Small Cell Lung Cancer, available at www.NCCN.org).

The choice of surgery for PM was previously controversial due to limited data from randomized controlled trials.[169-177] EPP is more likely to achieve macroscopic complete resection than P/D, but is also associated with increased morbidity and mortality.[169,172,178-182] Furthermore, the evidence supporting the clinical benefit of surgery was mainly derived from case studies or non-randomized trials.[179,183-185] However, there has been a paradigm shift in the treatment of PM based on data from randomized clinical trials, which indicated that a non-surgical approach with systemic

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therapy may lead to better survival outcomes than cytoreductive surgery for patients with resectable PM, regardless of surgical procedure.[6,186-188]

The MARS trial was a multicenter study based in the United Kingdom that was designed to assess the feasibility of randomizing 50 patients with resectable mesothelioma who were candidates for trimodal therapy within a 1-year period.[187] Patients received 3 cycles of platinum-based chemotherapy after initial staging and were randomized to receive either EPP or no EPP. The most common chemotherapy regimens used within the trial included cisplatin with gemcitabine and cisplatin with pemetrexed. Among the 16 patients who underwent EPP, only 8 received postoperative hemithoracic RT. There was high morbidity associated with EPP, as the 12-month survival rate was 52.2% in the EPP group and 73.1% in the no EPP group (HR, 1.9; *P* = .082). The median survival was 14.4 months in the EPP group and 19.5 months in the no EPP group. Perioperative deaths occurred in 12.5% (3/24) of the EPP group. The median quality-of-life scores were lower among patients who received EPP compared with those who did not. Due to the high morbidity rate in the EPP group and the low number of randomized patients overall, the authors concluded that a larger randomized trial was not feasible. Noted limitations of the MARS trial include low statistical power due to the small sample size, lack of standardized chemotherapy regimens, and poor protocol compliance.[189]

Data indicate that lung-sparing surgery also may not confer a survival benefit in patients with PM. MARS2 was a phase 3 multicenter randomized trial based in the United Kingdom that compared survival outcomes in patients with PM who received lung-sparing P/D (or extended P/D) in combination with chemotherapy versus those who received chemotherapy alone.[188,190] Patients who were deemed surgically resectable received 2 cycles of platinum chemotherapy with pemetrexed and were randomized to P/D (or extended P/D), followed by 2 or 4 more cycles of the same chemotherapy or chemotherapy alone (2 or 4 more cycles of platinum chemotherapy with pemetrexed). The median OS was 24.8 months in the chemotherapy only group compared with 19.3 months in the chemotherapy plus surgery group (HR, 1.28; *P* = .032), suggesting that there was a 28% increased risk of death among patients who received surgery.[188] The HR for OS between surgery plus chemotherapy versus chemotherapy alone was 1.12 among patients with epithelioid histology and 2.66 among those with non-epithelioid subtypes, demonstrating the negative impact of surgery on OS regardless of histology. Additionally, P/D with chemotherapy was also associated with more grade 3 or higher AEs, lower quality-of-life scores, and approximately two times higher health care costs than chemotherapy alone.

Given that routine cytoreductive surgery was linked with increased morbidity and mortality rather than a survival benefit in both randomized studies, systemic therapy is preferred over surgery as the primary treatment modality by the NCCN Panel for most patients with PM.

However, the Panel acknowledges that surgery remains an option for select individuals with clinical stage I PM with epithelioid histology and PS 0 to 2 following a careful surgical evaluation based on factors such as tumor histology, distribution, the patient's pulmonary reserve, and availability of adjuvant and intraoperative strategies. If the team determines that the patient may benefit from resection following surgical evaluation, then P/D is the recommended surgical option. EPP should only be selected in rare circumstances that require careful consideration of the total treatment plan that include both the patient and multidisciplinary team.[179,187,191,192]

Cytoreductive surgery is not recommended as a primary treatment option for patients with stage II to IV PM regardless of histology; systemic therapy is recommended for those with PS 0 to 2 (see *Systemic Therapy* in this Discussion for specific regimens). Prognosis with surgery (and other therapy) is substantially diminished in patients with N2 disease. Surgical

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resection should only be considered for patients with N2 disease at centers with expertise in PM or in a clinical trial.

Refer to the *Principles of Surgery* in the algorithm for additional guidance on surgical resection for PM.

Radiation Therapy

Patients with PM who have undergone P/D are candidates for RT if they have good PS, pulmonary function, and kidney function. However, it can be challenging to accurately deliver RT to the entire pleural surface without damaging radiosensitive sites, such as the lung and heart, especially when the lungs are intact.[193] Pleural intensity-modulated RT (IMRT), a technique more advanced than standard RT, has been evaluated in the context of PM to improve survival outcomes. A phase 3 randomized trial assessed postoperative radical hemithoracic RT (RHR) versus palliative RT given after lung-sparing surgery and chemotherapy in 108 patients with PM.[194] The median OS was longer with RHR than with palliative RT (25.6 months vs. 12.4 months; $P < .001$). The 2-year OS rate was 58% in the RHR arm versus 28% in the palliative RT arm (HR, 0.54; $P = .031$). In the RHR arm, 11 patients had grade ≥3 acute toxicity; 17 patients had grade 3 to 4 late toxicity. Nine patients experienced grade ≥2 pneumonitis.

A phase 2 trial (IMPRINT) evaluated the safety of hemithoracic pleural IMRT in patients with PM, given after chemotherapy (pemetrexed in combination with cisplatin or carboplatin) and P/D.[195] Patients in the study whose tumors were deemed unresectable after chemotherapy were eligible to receive pleural hemithoracic IMRT. Of the 27 patients who received pleural hemithoracic IMRT, grade ≥2 radiation pneumonitis was reported in 30%; improvement was reported with prednisone initiation. The 2-year OS was 59% in patients with resectable tumors and 25% in those with unresectable disease. Mediastinal nodal failure occurred in 22% (6/27) of patients; distant progression occurred in 48% (13/27) of patients. IMRT after systemic therapy and/or lung-sparing surgery has also been evaluated in other studies.[196-198]

Based on the results from these studies, the NCCN Panel recommends that pleural IMRT can be considered as part of primary treatment following systemic therapy (induction or adjuvant) with or without P/D in select patients with stage I PM with epithelioid histology. Given the technical difficulty associated with this treatment, the NCCN Panel emphasizes that IMRT should be performed in centers with expertise in this technique. The most appropriate timing of delivering RT (ie, after surgical intervention, with [or without] chemotherapy) should be discussed with a multidisciplinary team.

RT can also be used as palliative therapy for relief of chest pain, bronchial or esophageal obstruction, or other symptomatic sites associated with PM, such as metastases in bone or in the brain (refer to the NCCN Guidelines for Central Nervous System Cancers, available at www.NCCN.org).[69,79,199-201]

CT simulation–guided planning using either IMRT or conventional photon/electron RT is acceptable.[202-205] For treatment planning, PET scans can be used as indicated.[201] The clinical target volumes should be reviewed with the thoracic surgeon to ensure coverage of all the volumes at risk.

The dose of radiation should be based on the purpose of treatment. The total doses of radiation are described in the algorithm (see *Principles of Radiation Therapy*). The postoperative RT doses after P/D are 45 to 60 Gy in 1.8 to 2 Gy, with a higher dose to higher risk areas.[194,201,206] A dose of ≥60 Gy is recommended for macroscopic residual tumors, if the doses to normal adjacent structures are limited to their tolerances (see the NCCN Guidelines for Non-Small Cell Lung Cancer, available at

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www.NCCN.org).[201] The volume of postoperative radiation should cover the surgical bed within the thorax.[184,207-211] The optimal dose of RT for palliative purposes remains unclear. However, total doses of 20 to 40 Gy appear to be effective in providing relief from pain.[69,201,212,213]

Pleural IMRT is a highly conformal RT approach with improved coverage to the hemithorax at risk.[197,200,204] Advanced technologies may be used, such as image-guided RT (IGRT) for treatments involving IMRT, stereotactic radiosurgery (SRS), stereotactic body RT (SBRT), and intensity-modulated proton therapy (IMPT).[71,193,201,214,215] RT to the contralateral uninvolved lung should be minimized,[200,216,217] because fatal pneumonitis may occur with IMRT if strict limits are not applied.[218-220] The contralateral uninvolved mean lung dose should be kept as low as possible, preferably <8.5 Gy.[201,221] The volume of contralateral lung receiving low-dose RT (eg, 5 Gy) should be minimized.[194,222,223]

It has been controversial whether immediate (prophylactic) RT is useful for preventing instrument-tract recurrence after pleural intervention.[224-229] The phase 3 randomized SMART trial evaluated prophylactic radiotherapy versus deferred radiotherapy to assess the rate of recurrences in patients who had undergone pleural interventions for PM.[230] Patients in the deferred RT arm did not receive RT until procedure-tract metastases were evident. Among the 203 patients included in the intent-to-treat analysis, there was no significant difference in the proportion of those who experienced procedure-tract recurrence in the prophylactic RT arm (9%) versus the deferred RT arm (16%) (odds ratio [OR], 0.51; P =.14). Prophylactic RT also did not improve quality of life, decrease chest pain, or decrease the need for analgesic drugs. However, if patients did not receive chemotherapy, prophylactic RT did decrease the risk for procedure-tract metastases (OR, 0.16; P = .021). The NCCN Panel does not routinely recommend prophylactic RT to prevent instrument-tract recurrence after pleural intervention based on the SMART trial (see *Principles of Radiation Therapy* in the algorithm).

Treatment Recommendations for Pleural Mesothelioma

Treatment for patients diagnosed with PM is generally guided by clinical disease stage, histological subtype, and operability. Early referral to a multidisciplinary team with experience in mesothelioma is recommended. The following sections detail treatment recommendations for patients with good PS (0–2). Recommendations are stratified by clinical staging; for patients who undergo surgery, upstaging may occur. The Guidelines adhere to the American Joint Committee on Cancer (AJCC) 9th edition TNM staging system.[231]

For patients with PS 3 or 4, the NCCN Panel recommends best supportive care. RT and image-guided thermal ablation are palliative options for symptomatic pleural disease.[225,232,233] Refer to the *Principles of Radiation Therapy* and the *Principles of Supportive Care and Survivorship* in the algorithm for additional information.

Pleural effusion can cause discomfort for patients with PM. Thoracoscopic pleurodesis or placement of a drainage catheter can be used to manage pleural effusion.[86-90,94,234-236] Therapeutic/palliative thoracentesis can also be used to remove pleural fluid and thus decrease dyspnea either before treatment or for patients who are not candidates for more aggressive treatment.[69]

Clinical Stage I PM with Epithelioid Histology

NCCN Guidelines versions from prior years recommended multimodal therapy that consisted of surgery, systemic therapy, and/or RT as a primary treatment option for patients with clinical stage I to IIIA PM with epithelioid histology based on available clinical data.[202,237-239] Given the results from the MARS and MARS2 randomized trials (described in the

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Surgery section), which demonstrated that surgery with either EPP or P/D does not yield optimal survival outcomes for most patients with PM, the NCCN Panel has designated systemic therapy as a preferred option for patients diagnosed with clinical stage I PM and epithelioid histology (see PM-2 in the algorithm).[187,188] Pleural IMRT can also be considered for patients who are treated with systemic therapy.[195] However, the Guidelines recommend that pleural IMRT should only be performed in carefully selected patients at a center with experience with this technique.

Due to the risks associated with surgery, observation for progression may also be considered for patients with PS 0 to 2 who are asymptomatic with minimal burden of disease, if systemic therapy is planned when symptomatic or radiographic progression occurs.

Surgical evaluation for potential resection should be considered only for certain patients with clinical stage I PM and epithelioid histology who are medically operable and can tolerate the risks associated with surgery (see PM-2 in the algorithm). The NCCN Panel notes that the benefit of surgical resection in PM is unclear and there is no evidence that patient survival is improved with surgery when combined with systemic therapy versus systemic therapy alone. It is essential that patients receive a careful assessment that includes FDG-PET/CT and invasive mediastinal staging before surgery is performed (as described in the *Pretreatment Evaluation* section). If surgery is deemed a viable treatment option following surgical evaluation by the multidisciplinary team, then the NCCN Panel recommends P/D over EPP (see PM-3 in the algorithm). Additional factors to consider before treatment selection in this group of patients include the extent of disease burden, comorbidities, and patient preference.

Trimodal therapy comprised of chemotherapy, surgery, and RT has been assessed in patients with medically operable PM.[202,207,237,238,240-242] Chemotherapy in either the neoadjuvant or adjuvant setting are acceptable and result in similar outcomes.[239] In the phase 2 randomized EORTC 1205 trial, a total of 69 patients with borderline resectable T1–3 N0–1 mesothelioma were randomized to receive either induction chemotherapy, followed by surgery; or surgery, followed by adjuvant chemotherapy.[239] The chemotherapy regimen used in the trial was pemetrexed in combination with either cisplatin or carboplatin. The median OS was 27.1 months in patients who received surgery and adjuvant chemotherapy and 33.8 months among those who received induction chemotherapy followed by surgery.

The NCCN Panel recommends that systemic therapy can be given as either induction or adjuvant therapy if the patient will undergo surgery (see PM-3 in the algorithm). Studies assessing multimodal therapy for PM have generally included platinum-based chemotherapy regimens.[187,188,239] Data on the perioperative use of immunotherapy regimens for PM are limited; trials to evaluate the use of perioperative immunotherapy and chemoimmunotherapy are ongoing.[243]

Sequential pleural IMRT at a center with experience in this technique can also be considered after surgery and/or systemic therapy as it may improve survival outcomes.[195,198] Please refer to PM-3 in the algorithm and the *Radiation Therapy* section for additional information.

Systemic therapy is recommended by the NCCN Panel upon disease progression. Refer to the *Systemic Therapy* section for details on specific systemic therapy regimens.

Clinical Stage II-IV PM with Epithelioid Histology; Sarcomatoid or Biphasic Histology (Any Stage); or Medically Inoperable

The NCCN Panel currently recommends systemic therapy for patients with PM and PS 0 to 2 for those who are medically inoperable or refuse surgery, those with clinical stage II to IV PM (regardless of histology), or those with sarcomatoid or biphasic histology (regardless of clinical stage)

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(see PM-2 in the algorithm). Refer to the *Systemic Therapy* section for specific first-line and subsequent therapy regimens.

Observation for progression may be considered for patients with PS 0 to 2 who are not eligible for surgery and are asymptomatic with minimal burden of disease if systemic therapy is planned when progression occurs (either radiologic or symptomatic progression).

For patients with sarcomatoid or biphasic histology, surgery is associated with worse survival outcomes compared to patients with epithelioid histology.[9,244] Therefore, surgical resection is not recommended as an option for patients with non-epithelioid PM.

Prior versions of the Guidelines had recommended multimodal therapy with surgery, systemic therapy, and RT as an option for patients with clinical stage II to IIIA PM and epithelioid histology. However, given the increased mortality and morbidity associated with cytoreductive surgery versus no surgery reported in the MARS and MARS2 trials as described above,[187,188] the NCCN Panel no longer recommends surgery as a treatment option for patients with stage II to IIIA PM and epithelioid histology.

See PM-2 in the algorithm for complete treatment recommendations.

Summary

The NCCN Guidelines for Mesothelioma: Pleural provide recommendations for the initial evaluation and treatment of patients diagnosed with PM, a rare cancer originating from the mesothelial surfaces of the pleura. A multidisciplinary team with experience treating mesotheliomas should be involved in determining the optimal treatment strategy for each patient. Factors such as clinical stage of disease, medical operability, histological subtype, and PS should be taken into consideration. Systemic therapy is recommended as a primary treatment option for most patients with PM and PS 0 to 2. Recommended systemic therapy options include ICI- and/or chemotherapy-containing regimens. Observation may be considered after initial diagnosis if the patient is asymptomatic with minimal disease burden and systemic therapy is planned upon disease progression. Carefully selected patients with clinical stage I disease with epithelioid histology and good PS who can tolerate the risks associated with surgery may be candidates for multimodal therapy comprised of systemic therapy and surgery with or without RT. Best supportive care is recommended for patients with PS 3 or 4. The Guidelines will continue to be updated annually based on clinical evidence and consensus.

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